

Amaan Mohamed Kalemullah

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Education

- MS Data Science, Analytics and Engineering** Graduating May 2026
Arizona State University, Tempe, AZ
- **Coursework:** Statistical Machine Learning, Statistics for Data Analysts, Data Mining, Data Processing at Scale, Data Visualization **CGPA: 3.67/4.00**
- B.Tech Computer Science and Engineering** Aug 2020 – May 2024
Vellore Institute of Technology, Chennai, India
- **Coursework:** Design and Analysis of Algorithms, Deep Learning: Principles and Practices, Speech and Language Processing using Deep Learning, Discrete Mathematics and Graph Theory **CGPA: 8.34/10.00**

Technical Skills

Programming Languages and Databases: Python, SQL, R, Java, MySQL, PostgreSQL, MongoDB
ML & Statistical Methods: Predictive Modeling, Exploratory Data Analysis (EDA), Hypothesis Testing, Feature Engineering, Hyperparameter Tuning, Cross-Validation, Ensemble Methods, A/B Testing, Time Series Analysis, Statistical Modeling, Causal Inference, SHAP
Deep Learning: PyTorch, TensorFlow, Keras, CNNs, RNNs, Transformers, NLP, Computer Vision, HuggingFace
Big Data & Libraries: Pandas, NumPy, Scikit-learn, Apache Spark, PySpark, Matplotlib, Seaborn
MLOps & Cloud: FastAPI, Streamlit, MLflow, Docker, Git, AWS

Professional Experience

- Arizona State University, Tempe, AZ: Teaching Assistant – Principles of Programming (CSE340)** Oct 2025 – Dec 2025
- Strengthened systematic problem-solving skills directly applicable to ML pipeline development by supporting instruction of compiler theory concepts such as lexical analysis, type systems, and Lambda calculus, and evaluating **C programming** assignments covering data structures and algorithm design.
- NeuralHire, Wylie, TX (Remote): Machine Learning Intern** Jul 2023 – Oct 2023
- Improved behavioral analysis model responsiveness by optimizing a real-time face detection pipeline using **Haar-Cascade** classifiers in **OpenCV**, reducing off-screen tracking latency across multimedia research datasets.
 - Eliminated manual data handling from the video and speech processing workflow by engineering **10+ production Python scripts** (OpenCV, Praat), enabling scalable, reproducible **data pipelines** that accelerated downstream **model training** data preparation.

Projects

- Reinforcement Learning for EV Charging Optimization** Jun 2025 – Nov 2025
Python, PyTorch, Stable-Baselines3, Gymnasium, PySpark, PPO/SAC/TD3, Streamlit
- Reduced EV fleet peak demand by **15–25%** and costs by **10–20%** against a FIFO baseline by training **3 RL agents** (PPO, SAC, TD3) over **400,000 timesteps** in a custom **Gymnasium** environment using big data pipelines.
 - Identified **PPO** as the optimal model, achieving a **39% lower peak load**, and visualized training metrics and policy performance via a **Streamlit** dashboard.
 - Designed a multi-objective environment supporting dynamic pricing, stochastic arrivals, and V2G capabilities, utilizing **hyperparameter tuning** for stable convergence.
- Customer Churn Prediction** May 2025 - Aug 2025
Python, Scikit-learn, Random Forest, Gradient Boosting, Pandas, EDA, Feature Engineering
- Achieved **~80% accuracy** on the IBM Telco dataset by building an ML pipeline with **Random Forest** and **Gradient Boosting**, streamlining **ETL**, encoding, and feature selection.
 - Identified key churn drivers (contract type, tenure) using **EDA** and **SHAP** values, translating **model explainability** into actionable business retention strategies.
- AI-Powered Clinical Trial Outcome Predictor** Apr 2025 - Aug 2025
PyTorch, Random Forest, XGBoost, Gradient Boosting, OpenAI GPT, FastAPI, Streamlit, Scikit-learn
- Achieved **85% accuracy** and **94.6% AUC-ROC** by developing a **multi-modal pipeline** combining a PyTorch network, **ensemble models**, and GPT-based protocol analysis to predict clinical trial success.
 - Deployed an end-to-end ML system covering **EDA**, feature engineering, and **hyperparameter tuning**, backed by a **FastAPI** backend and **Streamlit** interface for real-time inference.

Publications

Kalemullah, A. M. (2024). *Deepfake Classification for Human Faces*. Proceedings of the 7th International Conference on Circuit Power and Computing Technologies (ICCPCT 2024), IEEE. DOI: [10.1109/ICCPCT61902.2024.10672973](https://doi.org/10.1109/ICCPCT61902.2024.10672973)